

ADVANCED COMPUTER NETWORKS
(MTITE101)

Submitted By: **Aniket Sonker**
Roll no: **2026PIT4227**
Branch: **Information Technology**
Submitted to: **Karan Gupta sir**

Practical 1:

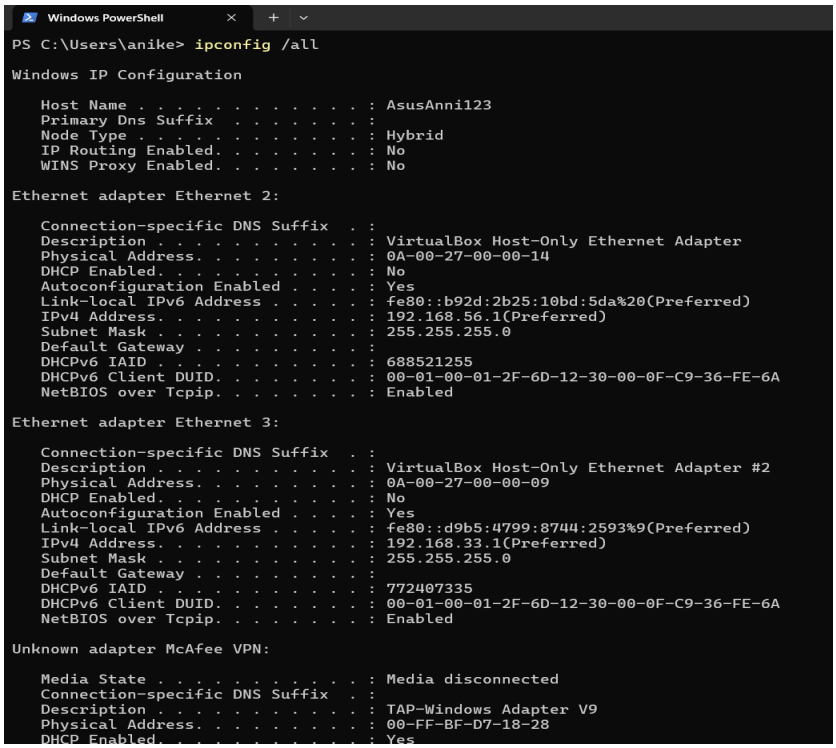
1. Aim: To study network configuration, routing tables, connectivity, packet paths, and IP-to-MAC address mapping using networking commands in the macOS Terminal.

2. Commands Run:

1: ipconfig /all

Function: Displays network configuration parameters including local IPv4/IPv6 addresses, MAC address, subnet mask, default gateway, and DNS servers.

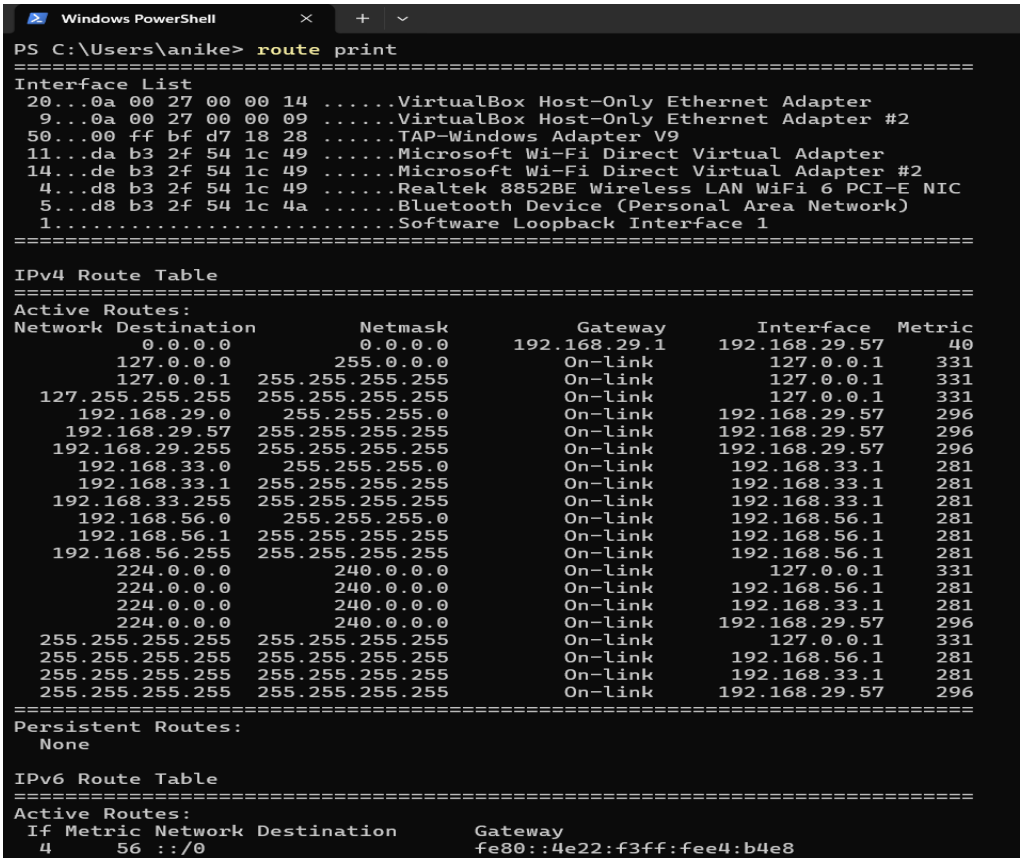
Output:



2: route print

Function: Displays the active IPv4 routing table, including network destinations, netmasks, gateways, active interfaces, and metrics.

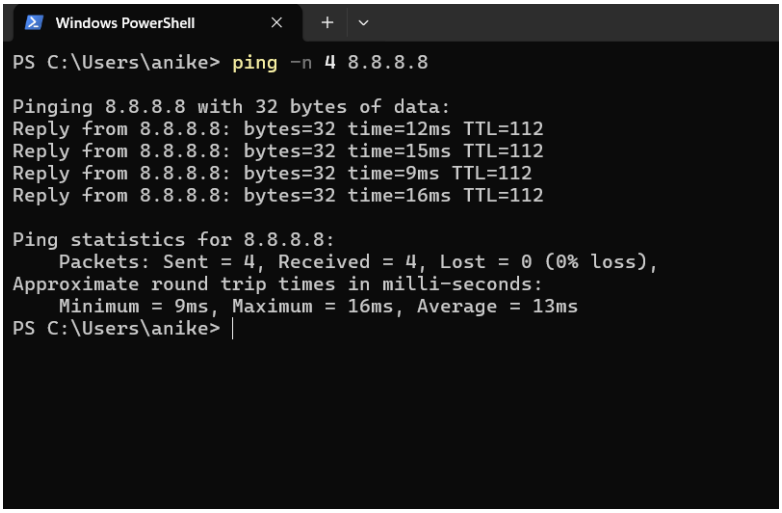
Output:



3: ping -n 4 8.8.8.8

Function: Sends ICMP Echo Request packets (4 requests specified by -n 4) to test connectivity, round-trip times, and packet loss.

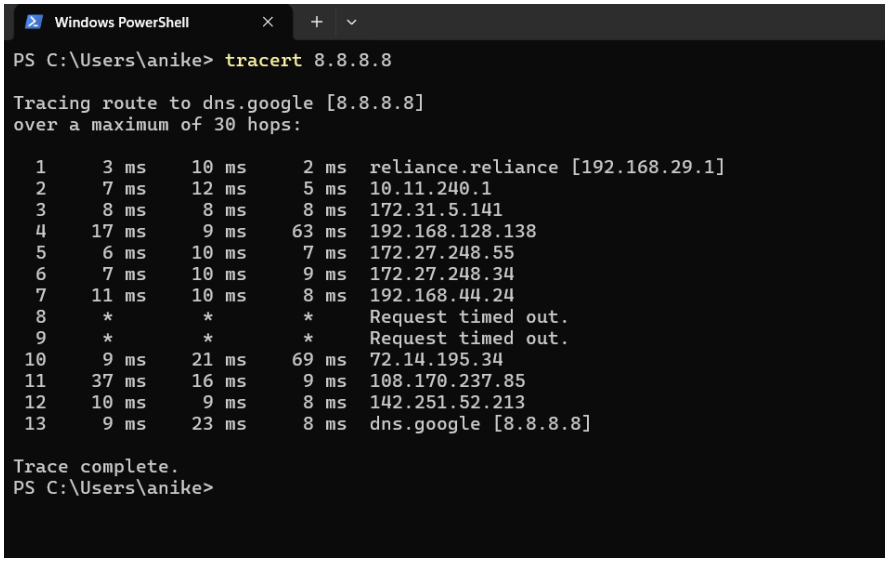
Output:



4: tracert 8.8.8.8

Function: Traces the path and measures latency across each intermediate router hop to reach the target IP.

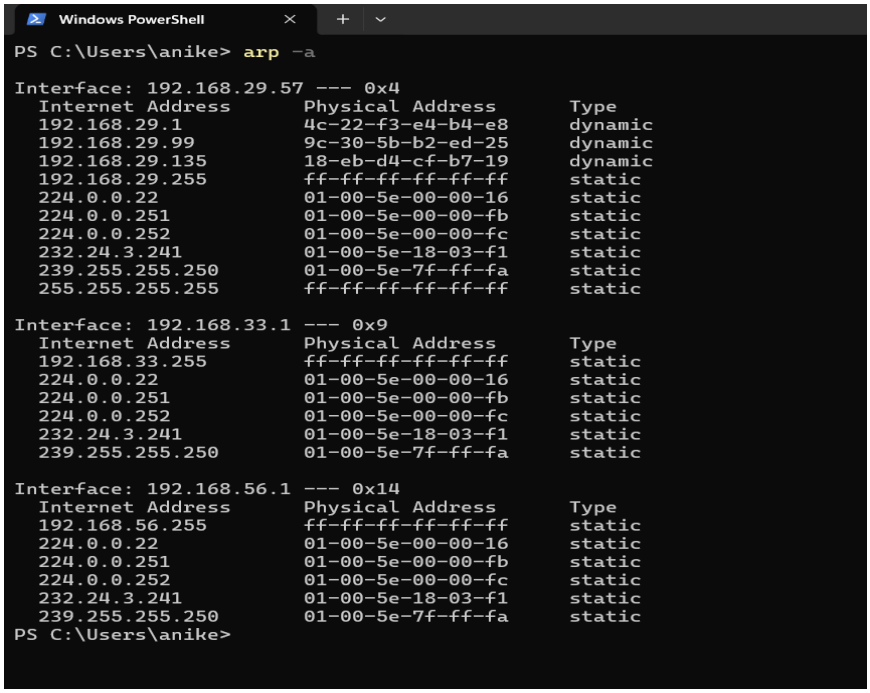
Output:



5: arp -a

Function: Displays the Address Resolution Protocol (ARP) cache, showing current IPv4-to-MAC address mappings on your local network.

Output:



3. Result: The networking commands were successfully executed, and the network interface configuration, IPv4 routing table, connectivity, network path, and ARP mappings were successfully observed and analyzed.